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12 October, 2023

ELEN 50 Lab 3

1.

	Rs Voltage (kOhm)	Rs Current (ohm)	R1 (ohm)	R2 (ohm)
Nominal Values	100 k-ohm	100 ohm	10k, 30k, 51k, 75k, 100k	51,100,270,430, 510
Measured Values	98.980 k-ohm	98.519 ohm	9.973k, 32.88k, 50.439k, 74.537k, 98.809	50.643 ,98.876, 267.43, 419.28, 500.39

2.

Measured R1 (kOhm)	Expected V1 (V)	Measured V1L (V)	Percentage Error E1 (%)	R_{vmext} (k-ohm)
9.973	0.4577	0.4573	-0.0873	10^7 * 1.1066
32.88	1.247	1.244	-0.2406	10^7 * 1.1057
50.439	1.689	1.682	-0.4144	10^7 * .9627
74.537	2.148	2.138	-0.4655	10^7 * .9248
98.809	2.498	2.484	-0.5604	10^7 * .8876

Question 4.

Rvmax avg: 10^7 * 0.99784 ohms = 9.9784 megaOhms

Percent Error= [(V1l-V1)/V1] *100

Question 5.

The voltmeter's resistance influences the circuit by acting as a resistor in parallel. Even if the resistance is really high to simulate infinite resistance, it still has a minor effect.

Part 2

Measured R2 (Ohm)	Expected i2 (A)	Measured i2L (A)	Percentage Error E1 (%)	R_{amext} (Ohm)
50.643	0.0134	0.01331	-0.6761	1.1010
98.876	0.0101	0.01007	-0.2970	1.2147
267.43	0.0055	0.00545	-0.9090	1.0235
419.28	0.0039	0.00385	-1.2825	1.6815
500.39	0.00334	0.00333	-0.001	1.6916

Percent Error= [(i2l-i2)/i2] *100
Ramext avg: 1.3425 ohms

Question 4:

It adds on to the total resistance in series, which is why the ammeter tries to be as low as possible to affect the circuit minimally.

Matlab scripts (just in case):

Part 1:
```
R1 = [9.973e3; 32.88e3; 50.439e3; 74.537e3; 98.809e3];
R2 = [51e3; 100e3; 270e3; 430e3; 510e3];
Rvm=100e3;
Ram=100e3;
Vs = 5;
Rs = 98.98e3;
```

```
V1 = R1.*(Vs./(R1+Rs))
V1L = [.4573; 1.244; 1.682; 2.138; 2.484];
E1 = (V1L-V1)./V1 * 100
Rvm2 = (R1.*-V1L.*Rs)./((R1.*V1L)-(R1.*Vs)+(Rs.*V1L))
```

Part 2:

```
R1 = [10e3; 30e3; 51e3; 75e3; 100e3];
R2 = [50.643; 98.876; 267.43; 419.28; 500.39];
Rvm=100e3;
Ram=100;
Vs = 2;
Rs = 98.519;
I2 = (Vs./(R2+Rs))
I2L = [.01331; .01007; .00545; .00385; .00333];
E2 = (I2L-I2)./I2 * 100
Ram2 = (Vs./I2L)-Rs-R2
```